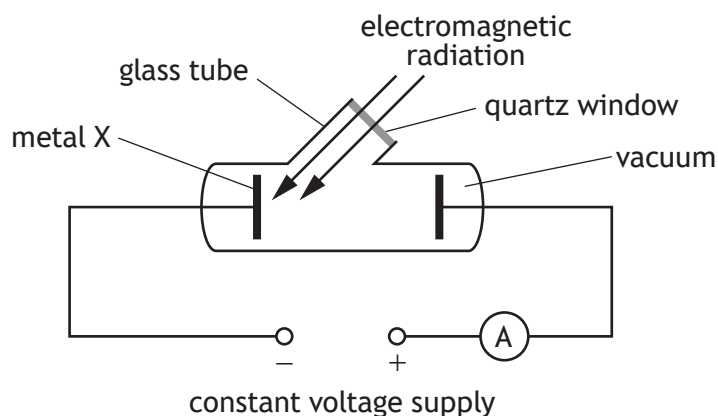
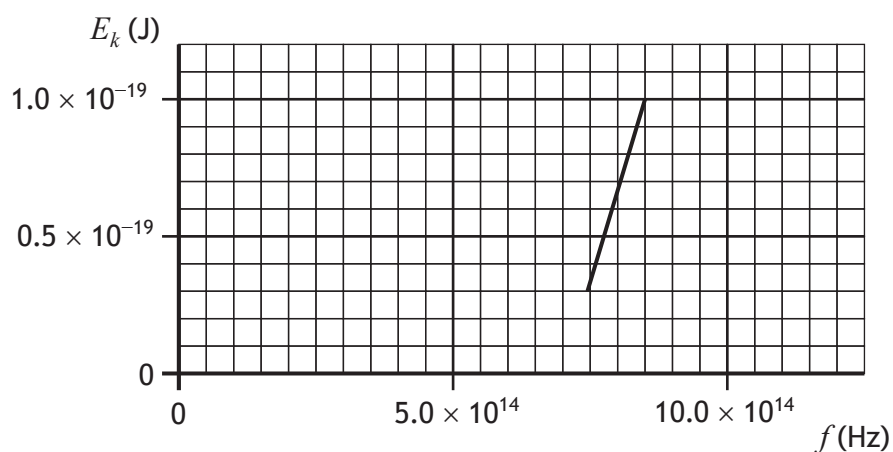


9. The apparatus shown is used to investigate photoemission.
Electromagnetic radiation is incident on metal X.



- (a) The frequency of the electromagnetic radiation is varied. The maximum kinetic energy of the photoelectrons emitted from metal X is determined for a range of frequencies.

The graph shows how the maximum kinetic energy E_k of the photoelectrons varies with frequency f .



Using the graph, determine the threshold frequency f_0 of metal X.

1

9. (continued)

(b) The work function of different metals is shown in the table.

Metal	Work function (J)
Potassium	3.7×10^{-19}
Calcium	4.6×10^{-19}
Zinc	5.8×10^{-19}
Gold	8.5×10^{-19}

Identify which of these metals is metal X.

Justify your answer by calculation.

Space for working and answer

4

